

Dadbeh Tavanaei

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Profile

Machine learning engineer and researcher with experience developing and evaluating deep learning systems in computer vision and NLP. Strong foundation in machine learning fundamentals, data preparation, experimentation, model evaluation, and error analysis. Experienced with Python and PyTorch and familiar with production-oriented development. Interested in building end-to-end AI systems from problem formulation and experimentation through deployment and iteration.

Education

Sharif University of Technology
M.Sc. in Computer Engineering

Starting October 2026

Sharif University of Technology November 2020 – February 2025
B.Sc. in Computer Engineering. GPA: 17.49/20.00
Thesis: Using semi-supervised training to enhance the interpretability of deep neural networks.

Technical Skills

Programming: Python, C++, C, Java
ML/DL: PyTorch, Transformers, Torchvision, Scikit-Learn
Data: SQL, NumPy, Pandas, Data Preprocessing, Feature Engineering, Statistical Analysis, Data Visualization
ML Engineering: Git, Docker, FastAPI, CI/CD fundamentals, Linux
Experimentation: Model Selection, Training, Evaluation, Error Analysis, Cross-Validation, Ensemble Methods
Languages: Persian (Native), English (TOEFL: 106)

Machine Learning Experience

Comma-Med Machine Learning Engineer June 2022 - September 2024

- Designed an ML data and labeling pipeline for mammogram lesion detection using a small set of accurately annotated images and a larger weakly labeled dataset.
- Developed and evaluated semi-supervised learning methods, including pseudo-labeling and consistency regularization, to improve lesion detection.
- Developed an ensemble segmentation model using multiple input resolutions for weakly supervised mammogram segmentation.
- Conducted controlled experiments across training strategies and model configurations and analyzed results to identify effective approaches and failure cases.

Sharif University of Technology, RIML Laboratory Research Assistant

Advisor: Prof. Rohban

- Developed and evaluated a deep learning approach for l_0 adversarial robustness using models pretrained against l_2 and l_∞ perturbations.
- Designed experiments measuring robustness, data efficiency, and computational requirements across different training strategies.

University of Nevada, Reno, Department of Computer Science and Engineering Research Assistant

Advisors: Prof. Bebis, Dr. Aliniya

- Implemented a GAN-based approach for medical image registration.
- Developed the experimental codebase and preprocessing pipeline for medical imaging datasets and evaluated multiple model configurations.

Projects

Transformer-Based Sentiment Analysis

- Fine-tuned a transformer-based language model on the Snappfood comments sentiment analysis dataset for Persian sentiment classification.
- Applied the fine-tuned model to analyze user comments on a medical website, classifying sentiment in patient reviews of doctors.

Tuberculosis X-Ray Segmentation

- Developed a U-Net-based binary segmentation model for tuberculosis-related lung segmentation using the Montgomery and Shenzhen chest X-ray datasets.
- Built the preprocessing pipeline and evaluated model performance on medical imaging segmentation tasks.

Deep Learning Projects

- Implemented and evaluated deep learning models using PyTorch, including CNN-based image classification, transfer learning, image captioning, conditional GANs, and clustering.
- Developed training workflows covering data preprocessing, model implementation, optimization, validation, and performance evaluation.

Teaching & Mentoring Experience

Allameh Helli 10, AI Olympiad

Spring 2026

AI Olympiad Instructor

- Taught artificial intelligence and machine learning concepts and guided students through practical problem-solving exercises.

Comma-Med

Fall 2023

Mentor, Lab Internship Program

- Mentored two interns in deep learning and computer vision, providing feedback on data preparation, model selection, training, and evaluation.
- Supervised a tuberculosis segmentation project from dataset preparation through model development and result analysis.

Sharif University of Technology, Artificial Intelligence

Fall 2023

Teaching Assistant

- Assisted in designing a practical assignment requiring students to implement a neural network from scratch, including forward propagation and backpropagation.

Sharif University of Technology, Probability and Statistics

Spring 2023

Teaching Assistant

- Assisted in developing theoretical assignments and solutions covering probability and statistical concepts used in machine learning and data analysis.